

Static Analysis of Protection Policies

KEY-SCOPE STATIC ANALYSIS | RESEARCH ADDENDUM | 25 SEPTEMBER 2026

Contribution and scope

KSSA v1 audits the experiment configuration before biometric data are loaded. It maps declared key scope to explicit findings and proposes a separate fresh-key comparison. This is a configuration analyzer, not a general source-code analyzer or a new biometric transform. Manish proposed this direction for discussion with Sani; professor approval and research priority are not established.

Algorithm

- 1. Parse the YAML scheme/condition matrix; reject unsupported required fields.
 - 2. Classify each pair: fresh-declared, shared across splits, shared projection, unknown.
 - 3. Emit reuse, correlation, disclosure and scheme-assumption findings.
 - 4. Block known policy risks; otherwise require review, never certify security.
 - 5. Propose fresh keys and hide slot labels; preserve all other experimental settings.
- Cost: $O(S \cdot C)$ scope classifications for S schemes and C conditions.

Finding	Decision / interpretation
KEY_REUSE	Block: hidden transforms recur across identity splits
KEY_CORRELATION	Block: a nonzero projection component is shared
SLOT_DISCLOSURE	Block: transform-group labels are exposed
Unknown / malformed	Block: unsupported policy cannot be certified
PolyProtect / Haar assumptions	Review: key freshness is insufficient evidence
Runtime assumptions	Review: secrecy, input norms and utility unverified

Prospective matched experiment

MOBIO and FEI; IoM-GRP and PolyProtect; two overlapping identity partitions; three model seeds; one and ten records. Compare pool 4 with the analyzer-generated fresh-key candidate using the same inputs, set seed, key seed, model seeds and training caps (120 epochs, patience 30). The mechanisms produce different key draws. All 16 cells / 96 trained endpoints completed in 352.77 seconds, within the 3,600-second budget.

Measured Reduction in Learned Linkage

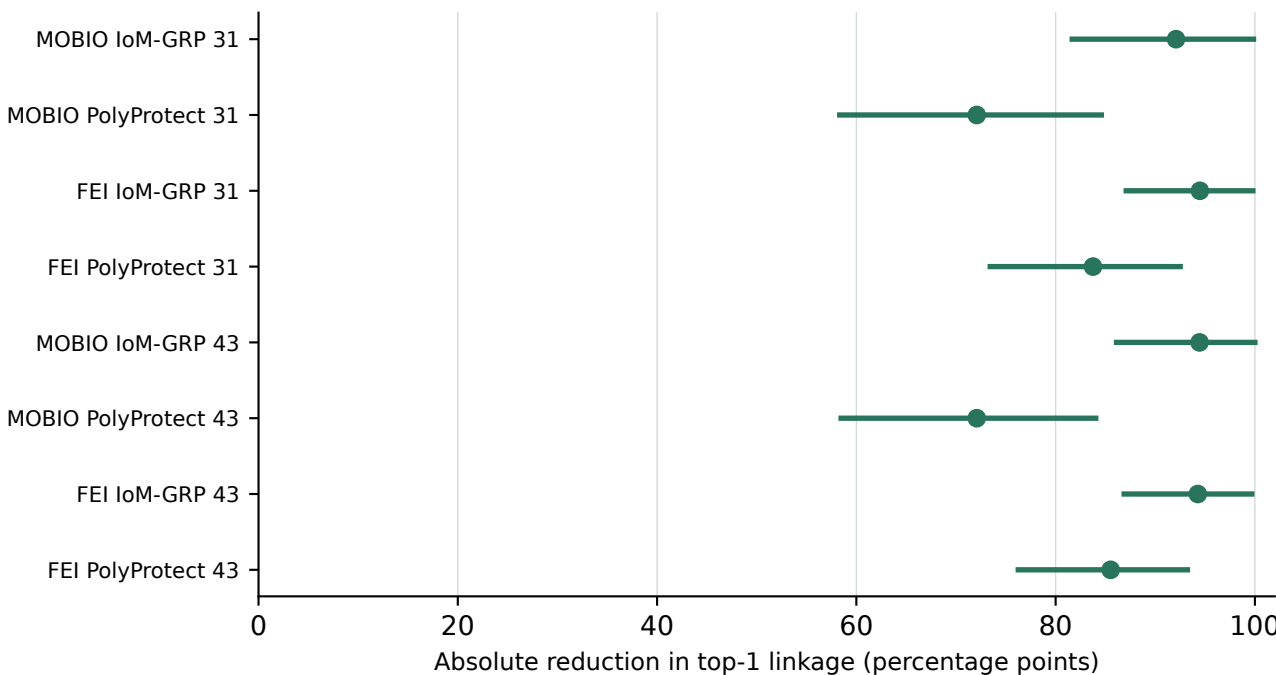
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Primary endpoint: pool 4 minus fresh, ten records

8/8 planned contrasts have positive paired 95% intervals and Holm-adjusted $p < .05$. Each row uses three newly trained mean-pool MLP seeds. Intervals resample model seeds and identity clusters; p-values use identity sign-flips of seed-mean differences. No security equivalence follows.

Data	Scheme	Split*	Before %	After %	Drop pp	95% CI (pp)
MOBIO	IoM-GRP	31	95.42	3.33	92.08	[81.66, 99.86]
MOBIO	PolyProtect	31	75.00	2.92	72.08	[58.33, 84.58]
FEI	IoM-GRP	31	97.08	2.60	94.48	[87.08, 99.79]
FEI	PolyProtect	31	85.62	1.88	83.75	[73.44, 92.50]
MOBIO	IoM-GRP	43	98.33	3.89	94.44	[86.11, 100.00]
MOBIO	PolyProtect	43	77.64	5.56	72.08	[58.47, 84.03]
FEI	IoM-GRP	43	96.77	2.50	94.27	[86.88, 99.69]
FEI	PolyProtect	43	87.71	2.19	85.52	[76.25, 93.23]

*Split suffixes 31/43 denote seeds 92531/92543. All eight Holm $p = 0.004$.



Chance: MOBIO 3.33%, FEI 2.50%. Every fresh learned CI includes chance; none proves equivalence.

Residual Risk and Reproducibility

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Native cross-record linkage is a different endpoint

Fresh PolyProtect still has descriptive native top-1 above chance in both datasets. This new native comparison is not null-tested or corrected. It supports retaining a review warning, not certifying unlinkability. These are cosine comparisons of protected gallery/probe records, not authorized authentication under compatible keys.

Dataset	Scheme	Split seed	Native top-1 %	Chance %
MOBIO	IoM-GRP	92531	5.45	3.33
FEI	IoM-GRP	92531	2.06	2.50
MOBIO	IoM-GRP	92543	4.55	3.33
FEI	IoM-GRP	92543	1.61	2.50
MOBIO	PolyProtect	92531	13.33	3.33
FEI	PolyProtect	92531	7.80	2.50
MOBIO	PolyProtect	92543	10.30	3.33
FEI	PolyProtect	92543	8.97	2.50

What this evaluation does not establish

One pool draw, overlapping partitions, three model seeds and small closed galleries limit inference. Baseline training uses paired access to the same hidden pool and known target grouping. Rules were motivated by earlier results, not evaluated on a blinded vulnerability corpus. No general detection accuracy, deployed authentication utility, new-transform superiority or universal privacy is claimed. SCface was not rerun because its local embeddings were unavailable.

Audit trail and use

Protocol: docs/protocols/static_policy_2026-09-25.md
Results: experiments/static_policy_2026-09-25/README.md
CLI: python -m biometrics_ai.protection.policy --config <policy.yaml>
--enforce: exit 2 for blocking findings; exit 3 for unresolved review.
Source/config/input hashes and software versions were recorded before execution. Public research seeds are not production secrets. Candidate policy changes require manual review and legitimate matching validation. Original studies and the September report are preserved.